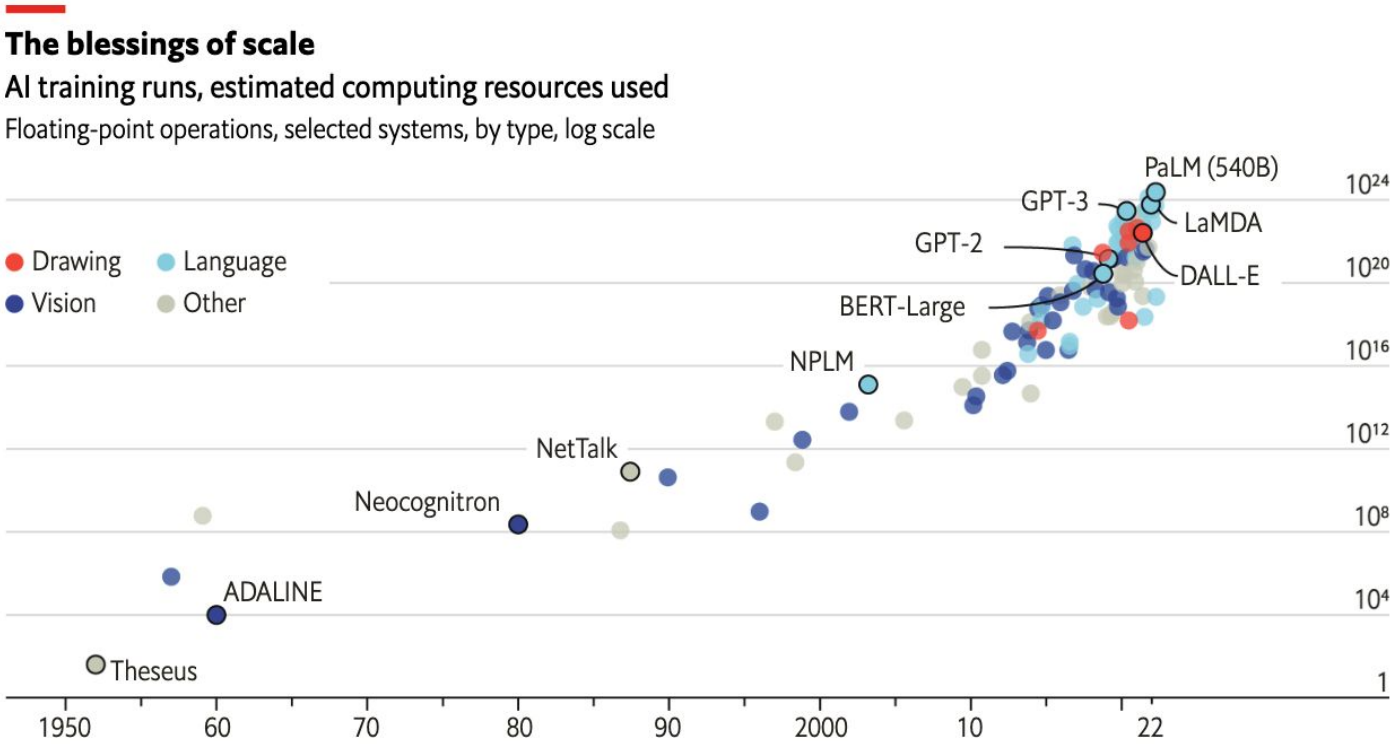
Why Ray

Machine learning is pervasive

Distributed computing is a necessity

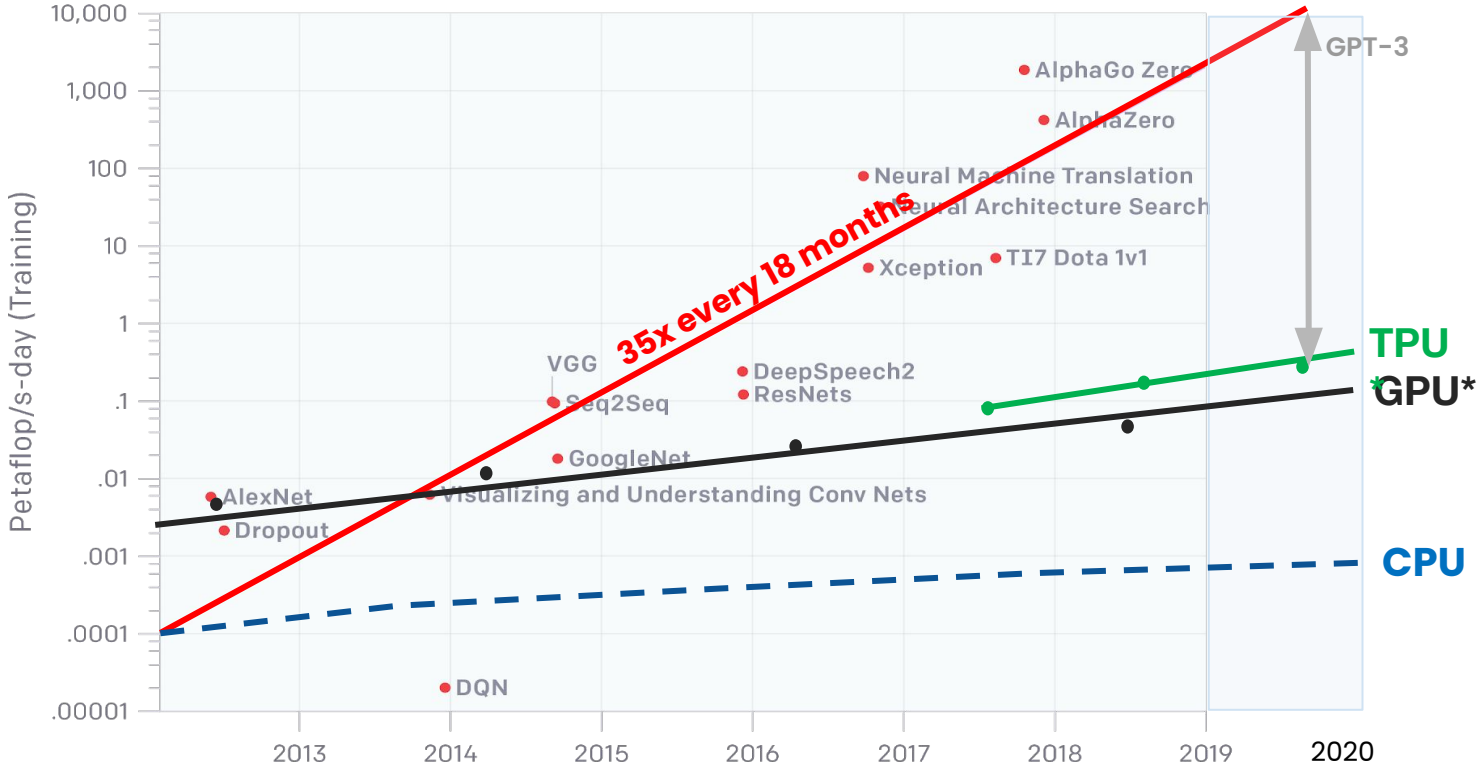
Python is the default language for DS/ML

Blessings of scale …



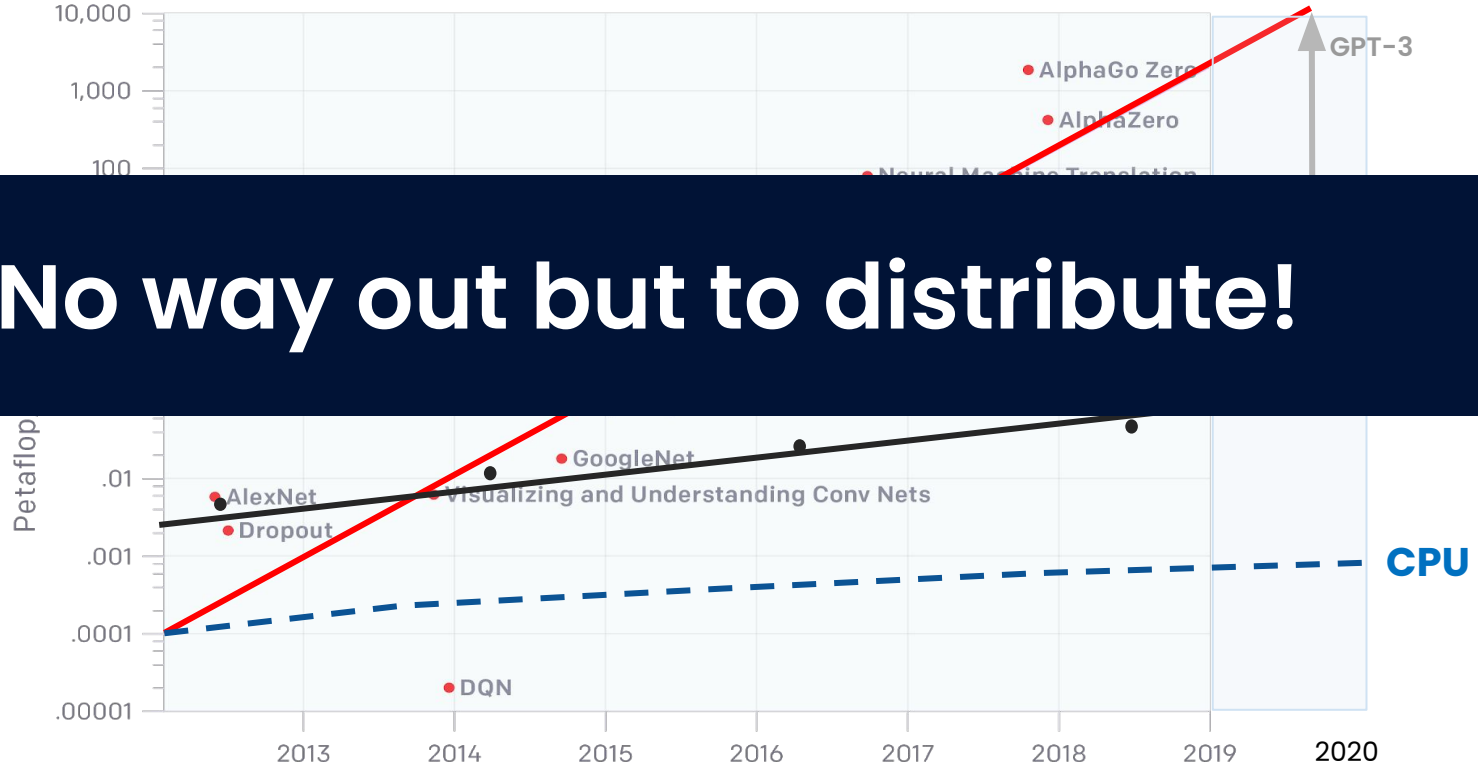
Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

Compute - supply demand problem



https://openai.com/blog/ai-and-compute/

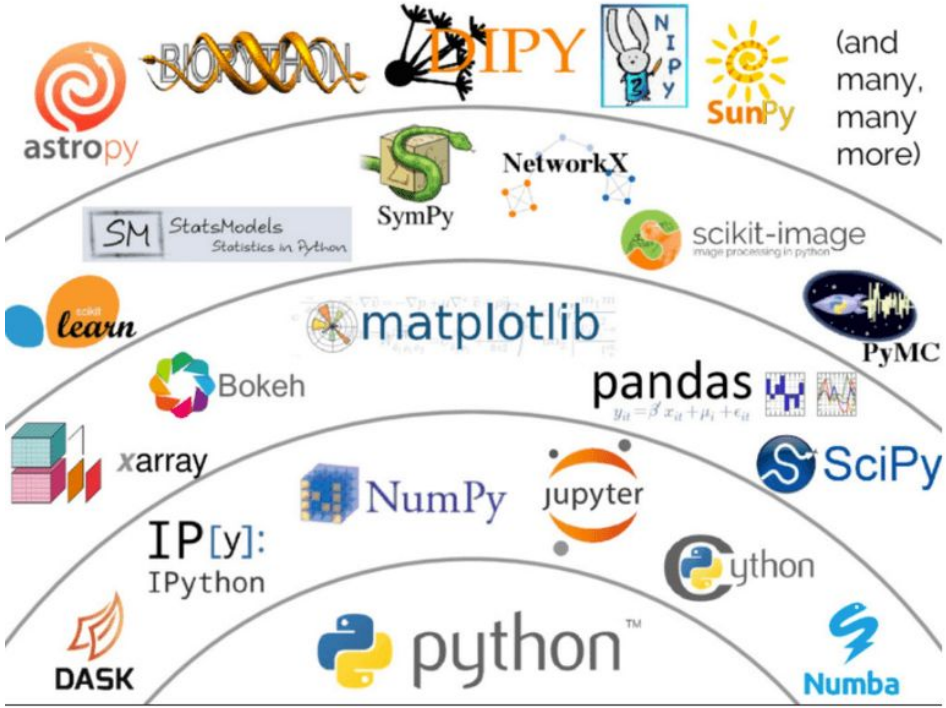
Specialized hardware is not enough



No way out but to distribute!

https://openai.com/blog/ai-and-compute/

Python data science ecosystem

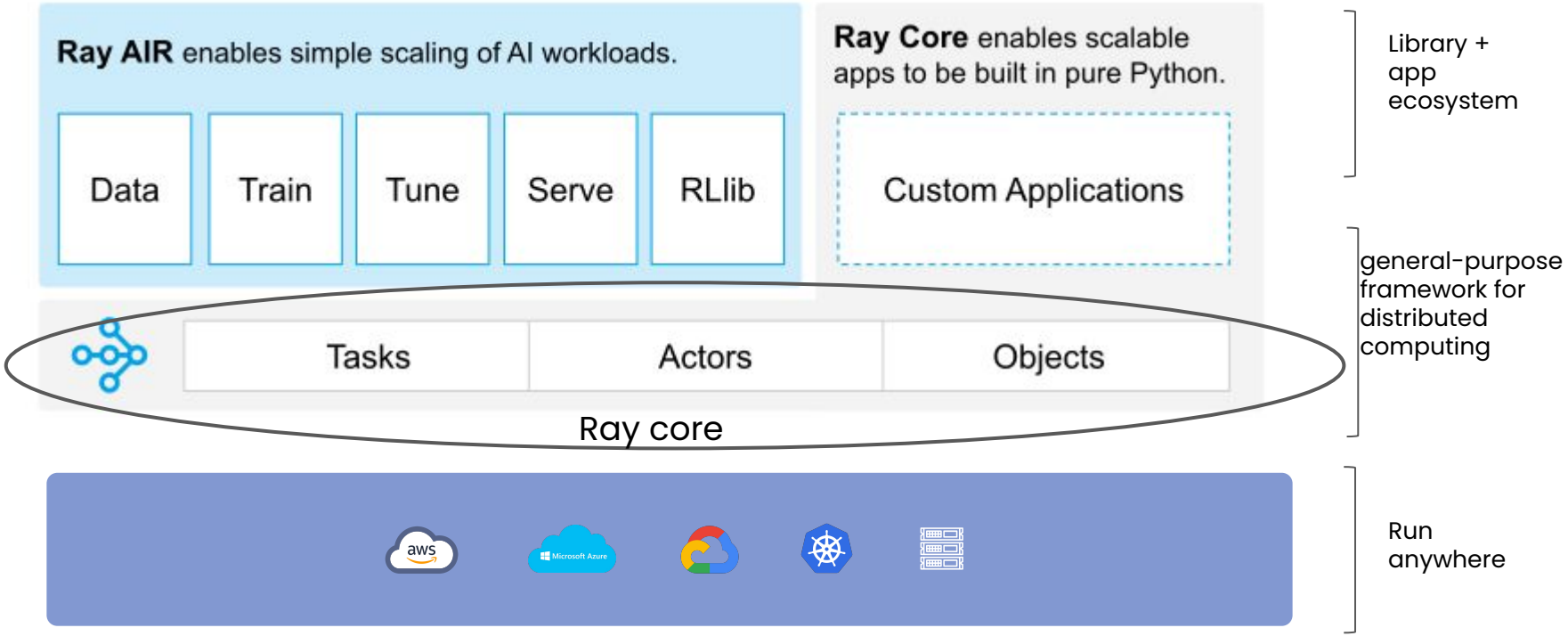


What is Ray

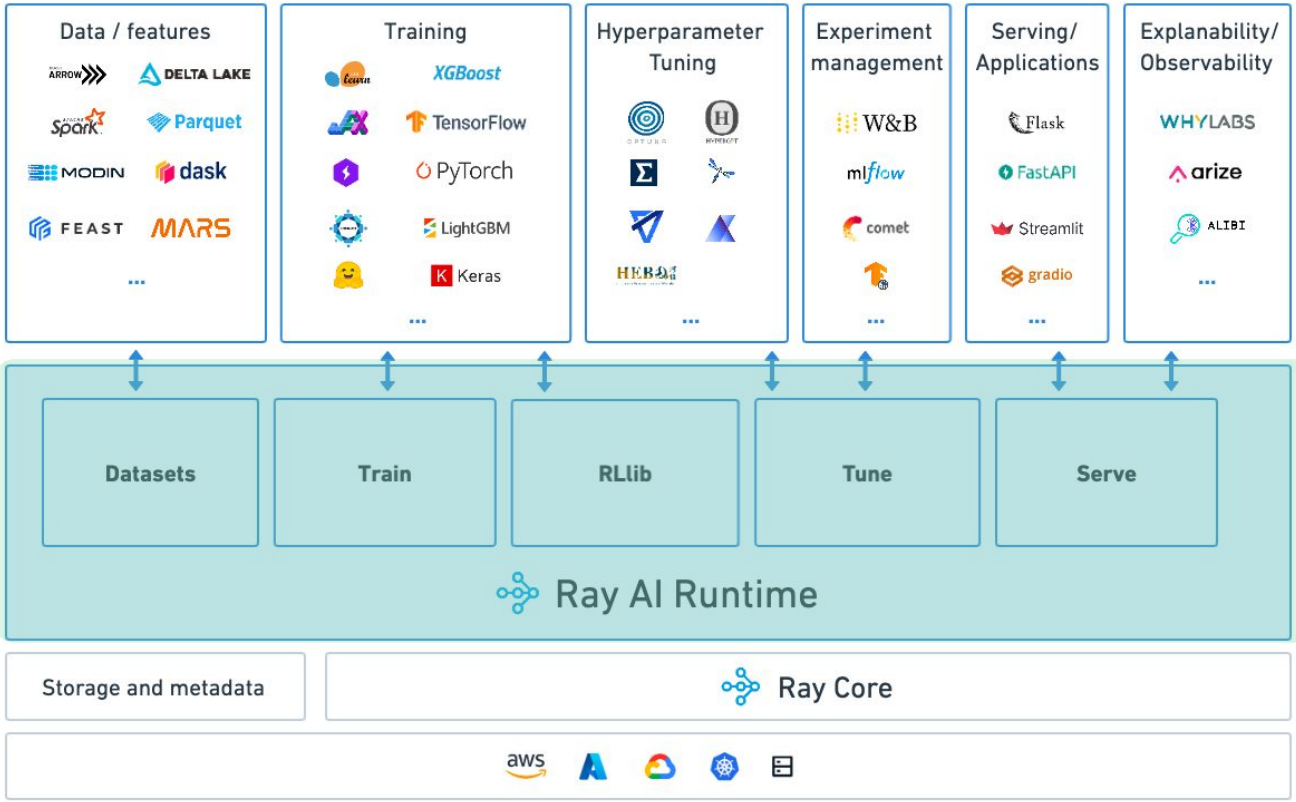
- A ***simple/general-purpose*** library for distributed computing
- An ecosystem of Python libraries (for scaling ML and more)
- Runs on laptop, public cloud, K8s, on-premise

A layered cake of functionality and capabilities for scaling ML workloads

A Layered Cake and Ecosystem

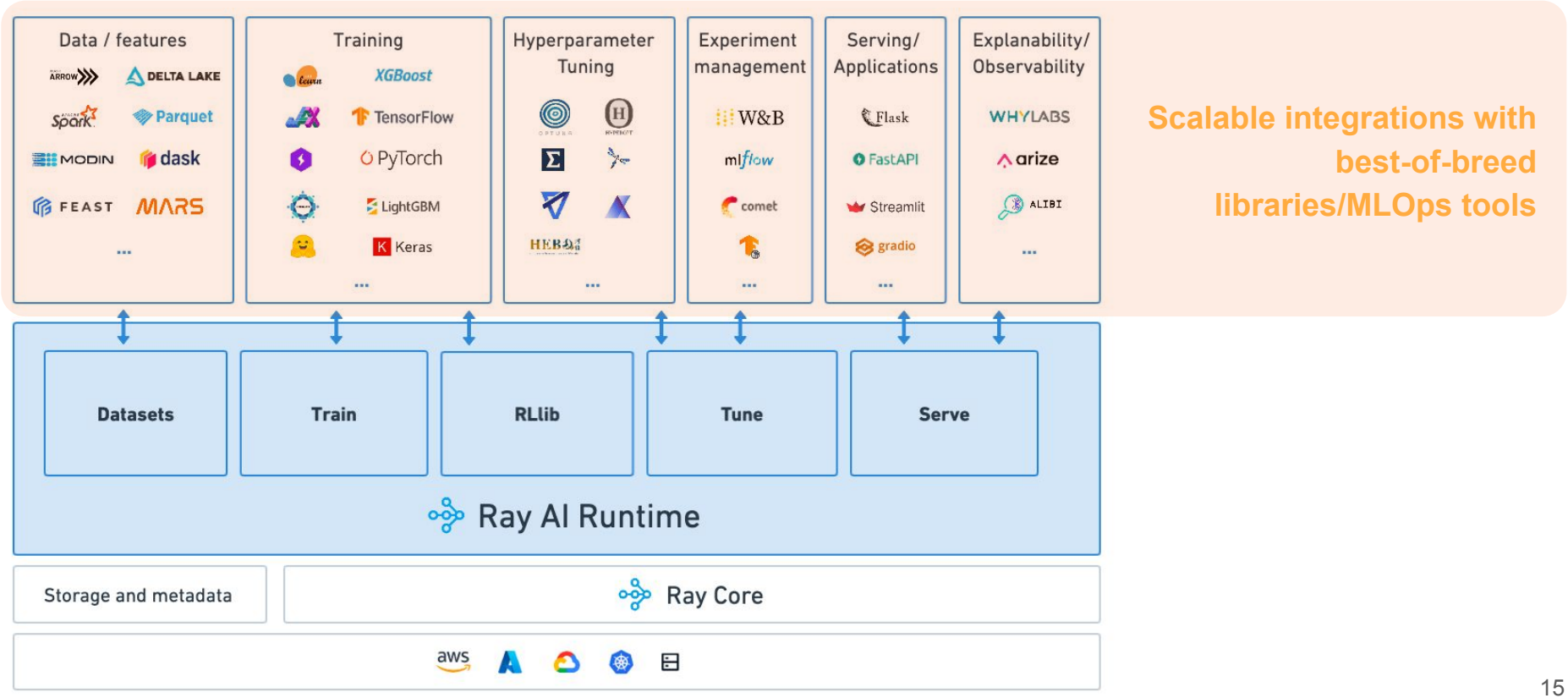


Ray AI Runtime (AIR) is a scalable runtime for end-to-end ML applications



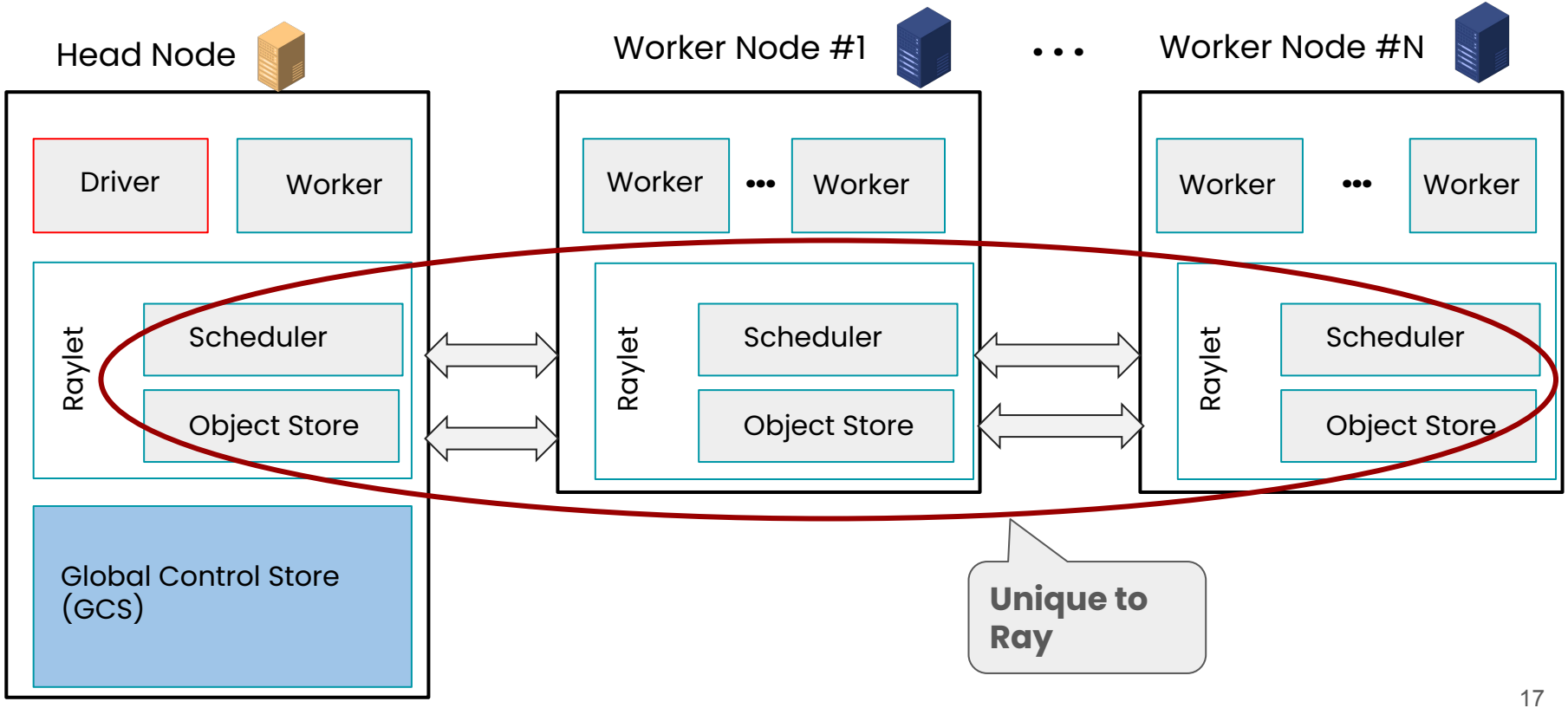
High-level libraries that make scaling easy for both data scientists and ML engineers.

Ray AI Runtime (AIR) is a scalable toolkit for end-to-end ML applications



Ray Architecture & Components

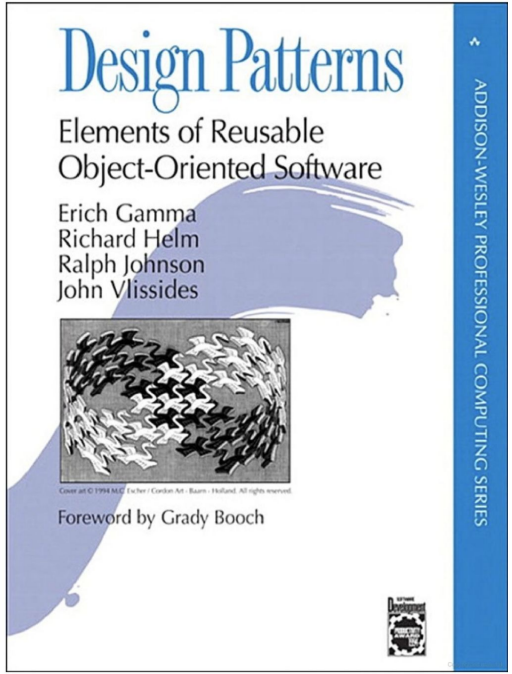
An anatomy of a Ray cluster



Ray distributed design & scaling patterns & APIs

Ray Basic Design Patterns

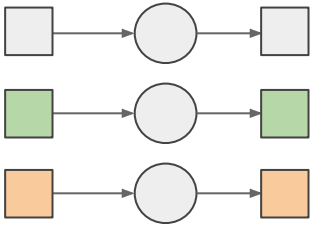
- Ray Parallel Tasks
  - Functions as stateless units of execution
  - Functions distributed across the cluster as tasks
- Ray Objects as Futures
  - Distributed (immutable objects) store in the cluster
  - Fetched when materialized
  - Enable massive asynchronous parallelism
- Ray Actors
  - Stateful service on a cluster
  - Enable Message passing



1. [Patterns for Parallel Programming]
2. [Ray Design Patterns]
3. [Ray Distributed Library Integration Patterns]

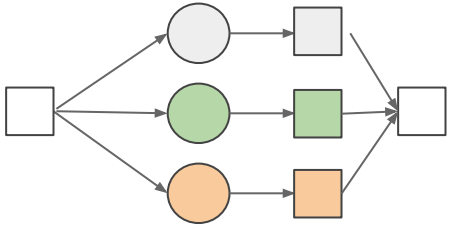
Scaling Design Patterns

Batch Training / Inference



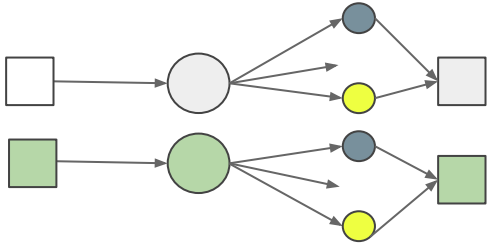
Different data / Same function

AutoML



Same data / Different function

Batch Tuning



Different data / Same function /

Compute

Data

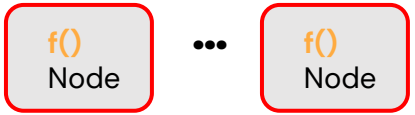
Python → Ray APIs

```
def f(x):
   # do something with
x:
   y = ...
   return y
```

Task →

```
@ray.remote
def f(x):
   # do something with
x:
   Y = ...
   return y
f.remote()
```

Distributed →




```
class Cls():
  def
__init__(self, x):
  def f(self, a):
        ...
  def g(self, a):
        ...
```

Actor →

```
@ray.remote
class Cls():
  def __init__(self,
x):
  def f(self, a):
        ...
  def g(self, a):
        ...
cls = Cls.remote()
cls.f.remote(a)
```

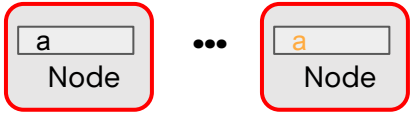
Distributed →

```
import numpy as np
a= np.arange(1, 10e6)
b = a * 2
```

Distributed immutable object →

```
import numpy as np
a = np.arange(1, 10e6)
obj_a = ray.put(a)
b = ray.get(obj_a) * 2
```

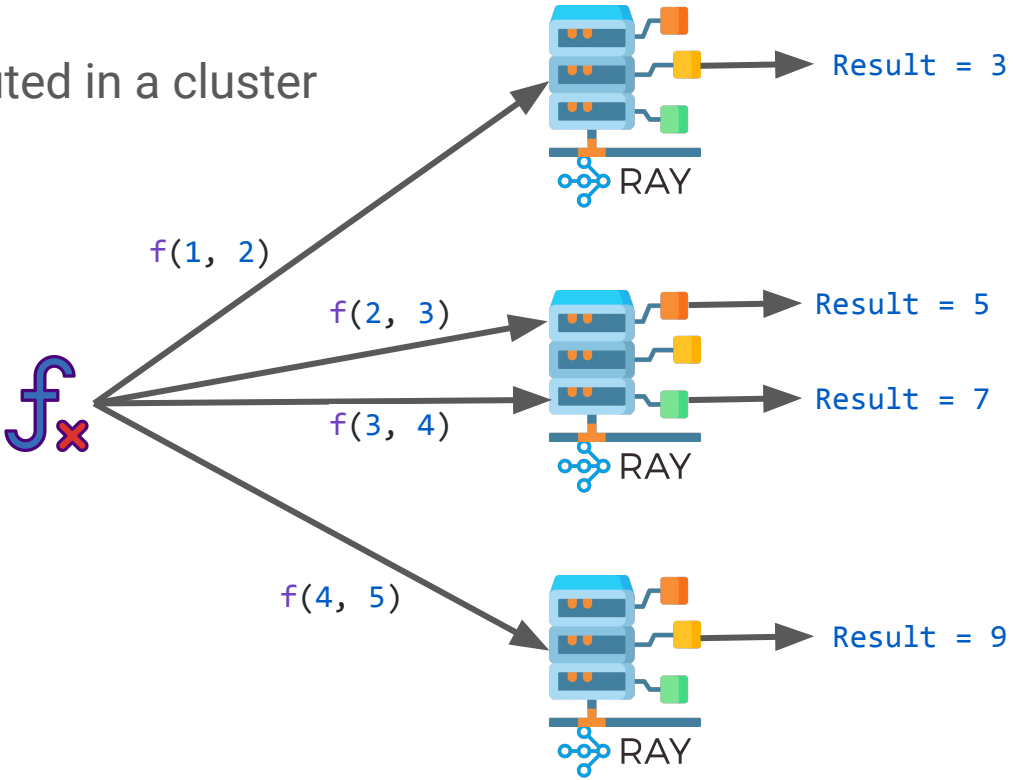
Distributed →



Ray Task

A function remotely executed in a cluster

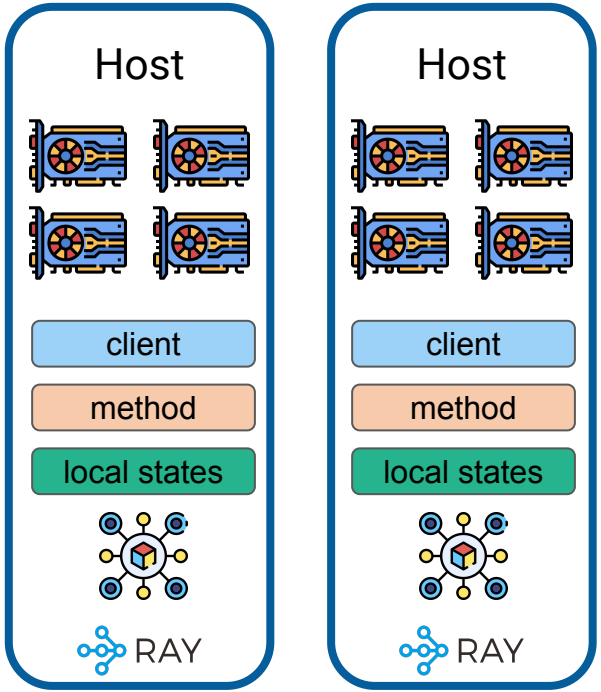
```
@ray.remote(num_cpus=2)
def f(a, b):
    return a + b

f.remote(1, 2) # returns 3
```



Ray Actor

A class remotely executed in a cluster

```
@ray.remote(num_gpus=4)
class HostActor:
    def __init__(self):
        self.num_devices = os.environ["CUDA_VISIBLE_DEVICES"]

    def f(self, output):
        return f"{output} {self.num_devices}"

actor = HostActor.remote()  # Create an actor
actor.f.remote("hi")  # returns "hi 0,1,2,3"
```



Function → Task

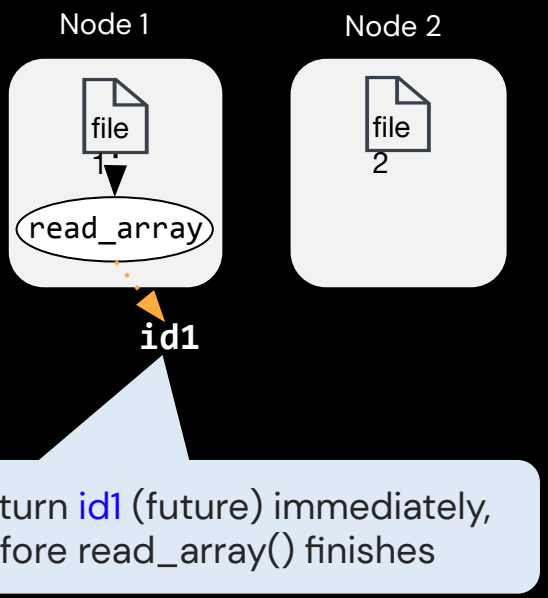
```
@ray.remote
def read_array(file):
    # read ndarray “a”
    # from “file”
    return a

@ray.remote
def add(a, b):
    return np.add(a, b)

id1 = read_array.remote(file1)
id2 = read_array.remote(file2)
id = add.remote(id1, id2)
sum = ray.get(id)
```

Class → Actor

```
@ray.remote(num_gpus=1)
class Counter(object):
    def __init__(self):
        self.value = 0
    def inc(self):
        self.value += 1
        return self.value

c = Counter.remote()
id4 = c.inc.remote()
id5 = c.inc.remote()
```

Task API

```
@ray.remote
def read_array(file):
    # read ndarray "a"
    # from "file"
    return a

@ray.remote
def add(a, b):
    return np.add(a, b)

id1 = read_array.remote(file1)
id2 = read_array.remote(file2)
id = add.remote(id1, id2)
sum = ray.get(id)
```



Task API

```
@ray.remote
def read_array(file):
    # read ndarray "a"
    # from "file"
    return a

@ray.remote
def add(a, b):
    return np.add(a, b)

id1 = read_array.remote(file1)
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sum = ray.get(id)
```
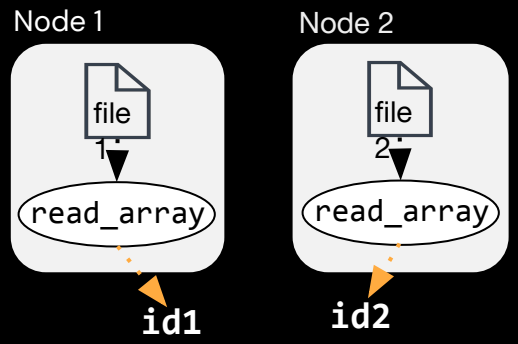
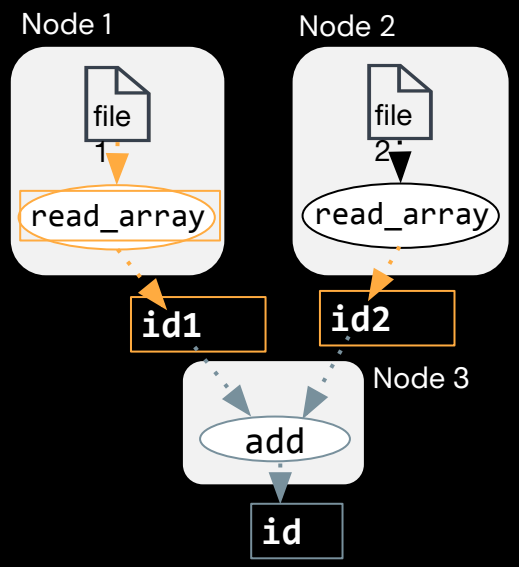



Dynamic task graph: build at runtime

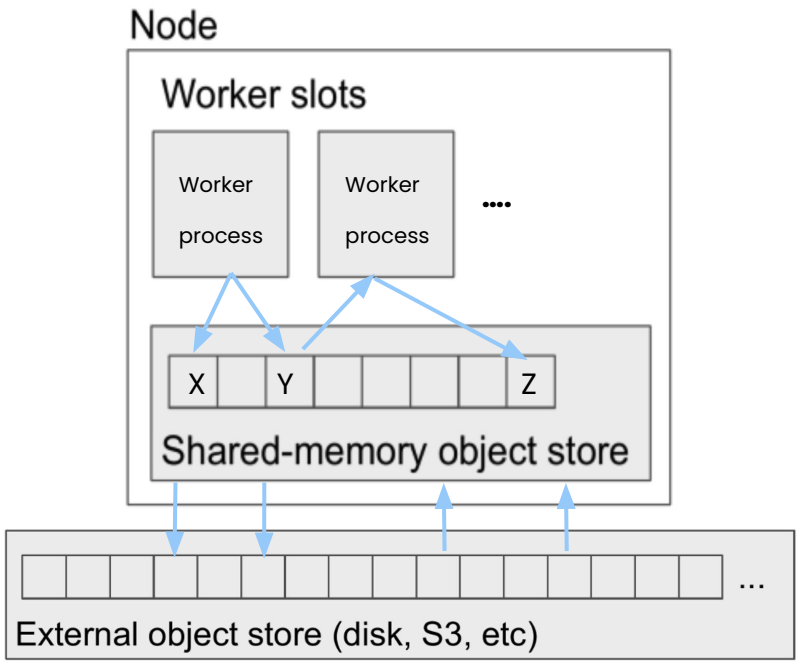
Task API

```
@ray.remote
def read_array(file):
    # read ndarray "a"
    # from "file"
    return a

@ray.remote
def add(a, b):
    return np.add(a, b)

id1 = read_array.remote(file1)
id2 = read_array.remote(file2)
id = add.remote(id1, id2)
sum = ray.get(id)
```

ray.get() block until result available

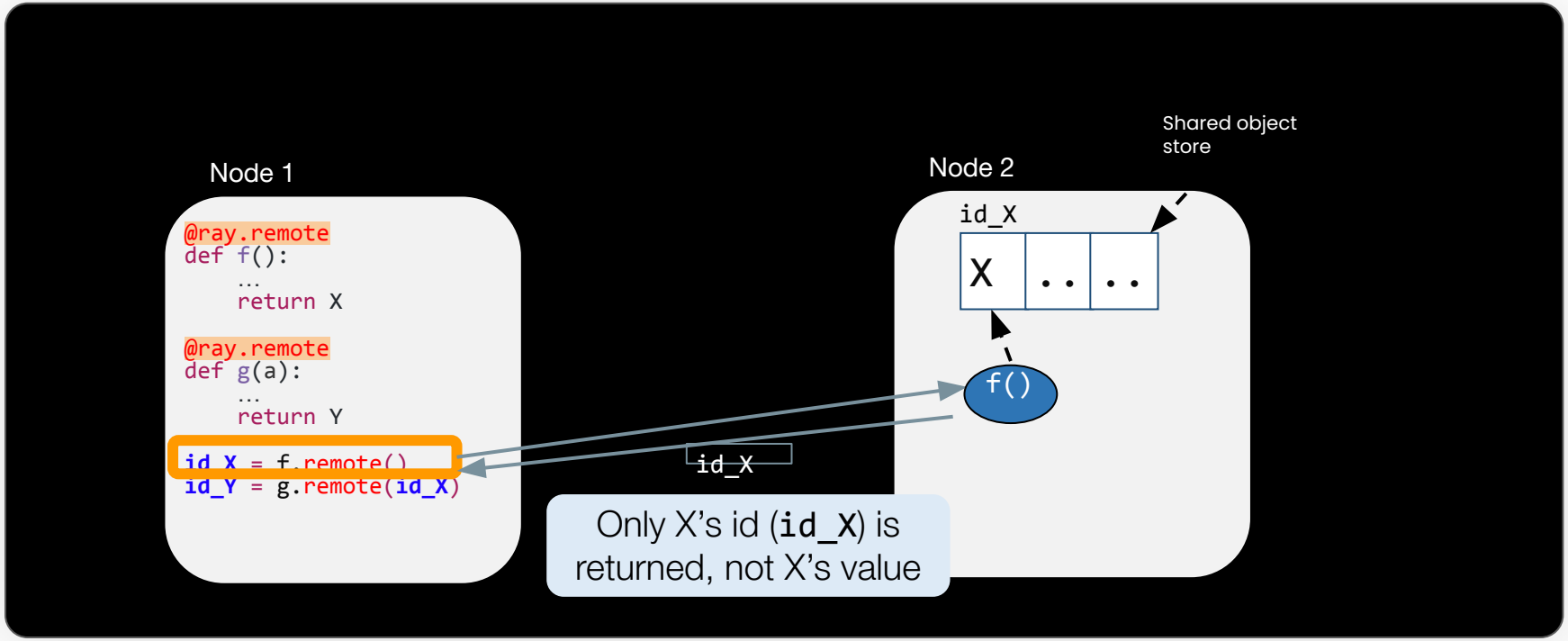


Distributed Immutable object store

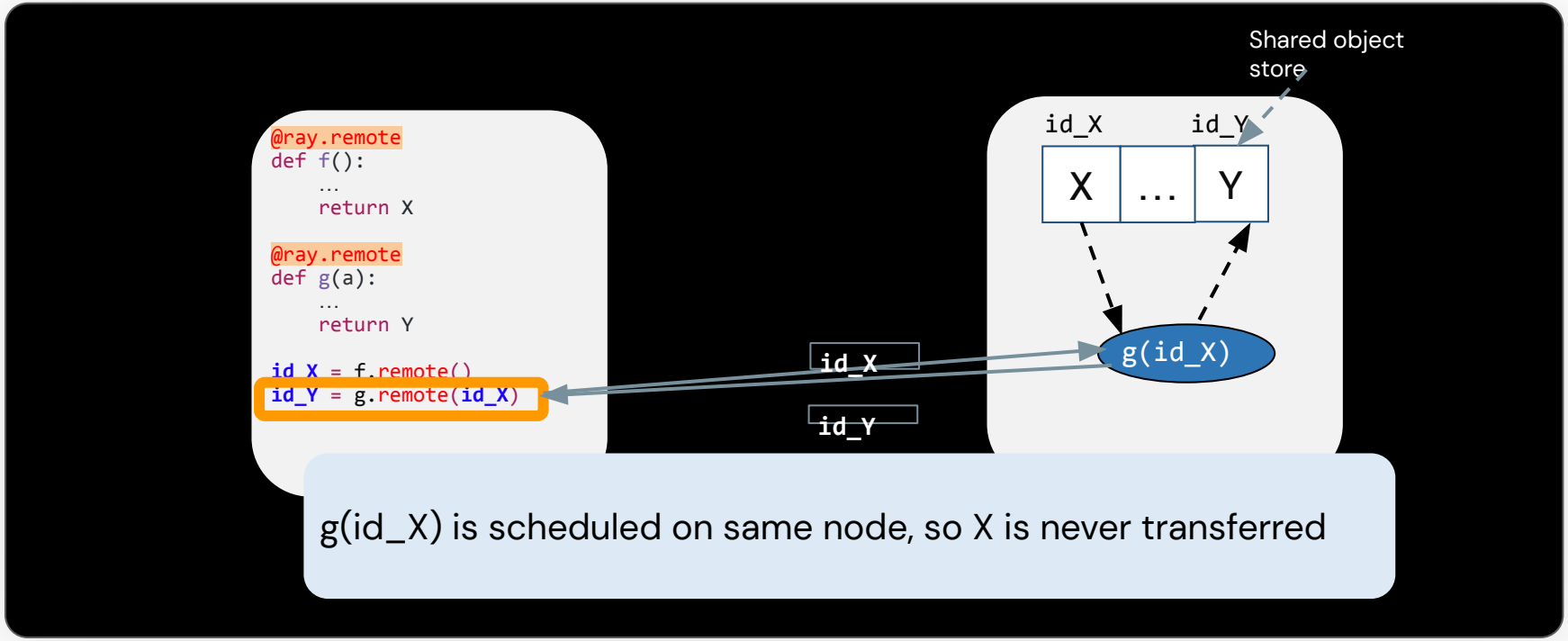


Spill over to external storage

Distributed object store



Distributed object store



Examples of Distributed Applications with Ray

Distributed Applications with Ray

ML Libraries

- Ray AI Runtime
- Distributed scikit-learn/Joblib
- Distributed XGBoost on Ray
- Ray Multiprocess Pool

All using Ray core APIs & patterns

Experimenting & Monitoring Services

- WhyLabs
- Arize AI
- W & B
- MLflow

All using Ray core APIs & patterns

ML Platforms & Integrations

- Merlin (Shopify)
- Meta (TorchX)
- AirFlow, Prefect
- HuggingFace
- Ludwig AI
- Uber
- Instacart
- Spotify

All using Ray core APIs & patterns

Ray Ecosystem: https://docs.ray.io/en/latest/ray-overview/ray-libraries.html

Generative AI, LLMs & Ray

Current state of the world..

LLM companies provide commercial APIs
(note: co:here and OpenAI both use Ray internally)

co:here

OpenAI

ANTHROP\C

Hugging Face

stability.ai

anyscale

Options for poor mortals …

1. Download the models
   HuggingFace
2. Fine-tune the models
   ???
3. Serve these models
   ???

anyscale

What are pain points in training/serving LLMs?

01	**Scaling is costly and hard to manage**	• Need spot instance support • Hard to run distributed workloads • Hard to optimize CPUs/GPUs
02	**Existing serving / inference solutions don't work**	• Individual replicas can't be distributed • Need to be able to integrate business logic
03	**Distributed Training hard to get working right**	• Hyperparameters need to be tuned • Need a platform to iterate very quickly at scale

anyscale

Ray provides generic platform for LLMs

01	**Simplify orchestration and scaling**	• Spot instance support for data parallel training • Easily spin up and run distributed workloads on any cloud • Optimize CPUs/GPUs by pipelining w/ Ray Data
02	**Inference and serving**	• Ability to support complex pipelines integrating business logic • Ability to support multiple node serving
03	**Training**	• Integrates distributed training with distributed hyperparameter tuning w/ ML frameworks

anyscale

What’s the LLM stack for Generative AI?

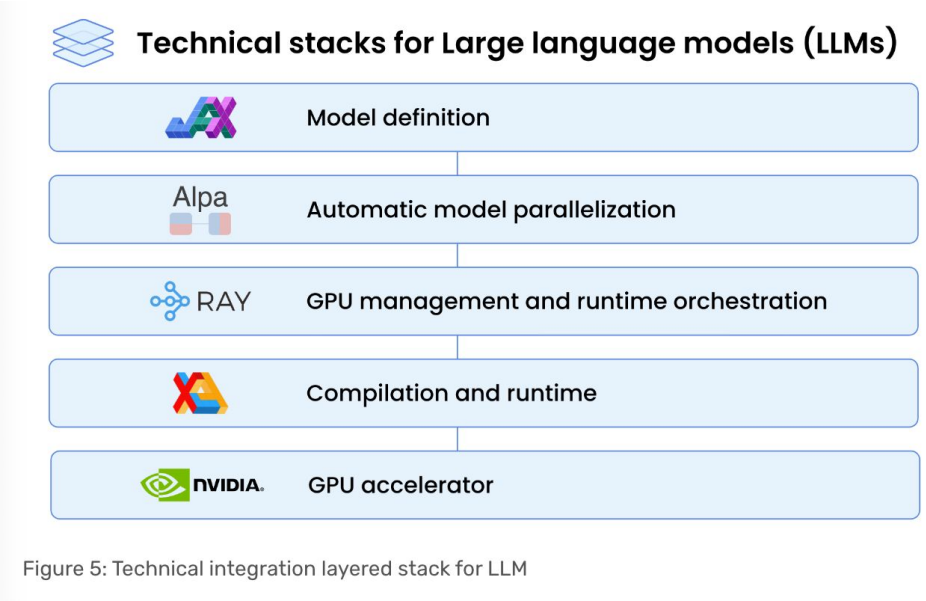


Figure 5: Technical integration layered stack for LLM

- HuggingFace for models
- DeepSpeed for optimized Training
- PyTorch for Framework
- Ray for Orchestration
- GPU or other hardware

What about Generative AI?

How Ray solves common production challenges for generative AI infrastructure

By Antoni Baum, Eric Liang, Jun Gong, Kai Fricke and Richard Liaw | March 20, 2023

This is part 1 of our generative AI blog series. In this post, we talk about how to use Ray to productionize common generative model workloads. An upcoming blog will deep dive into why projects like Alpa are using Ray to scale large models.

Training 175B Parameter Language Models at 1000 GPU scale with Alpa and Ray

By Jiao Dong, Hao Zhang, Lianmin Zheng, Jun Gong, Jules S. Damji and Phi Nguyen | March 22, 2023

This is part 2 of our generative AI blog series. Here we cover how Ray empowers large language models (LLM) frameworks such as Alpa. To learn how to use Ray to productionize generative model workloads, see part 1.

https://anyscale.com/blog

Faster stable diffusion fine-tuning with Ray AIR

By Kai Fricke | March 28, 2023

This is part 3 of our generative AI blog series that dives into a concrete example of how you can use Ray to scale the training of generative AI models. To learn more using Ray to productionize generative model workloads, see part 1. To learn about how Ray empowers LLM frameworks such as Alpa, see part 2.

How to fine tune and serve LLMs simply, quickly and cost effectively using Ray + DeepSpeed + HuggingFace

By Waleed Kadous, Jun Gong, Antoni Baum and Richard Liaw | April 10, 2023

This is part 4 of our blog series on Generative AI. In the previous blog posts we explained why Ray is a sound platform for Generative AI, we showed how it can push the performance limits, and how you can use Ray for stable diffusion.

Key Takeaways

- Distributed computing is a necessity & norm
- Ray’s vision: make distributed computing simple
  - Don’t have to be distributed programming expert
- Build your own disruptive apps & libraries with Ray
- Scale your ML workloads with Ray libraries (Ray AIR)
- Ray offers the compute substrate for Generative AI workloads